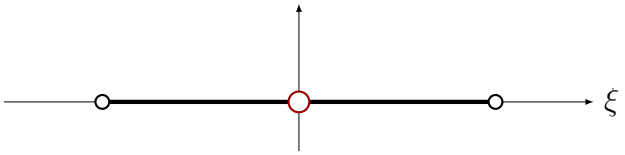
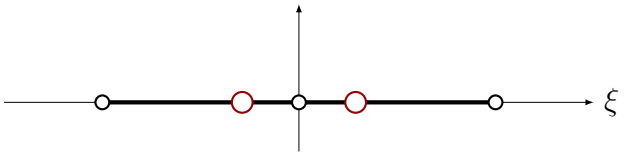
Gauss-Legendre points and weights in 1D

Order 1



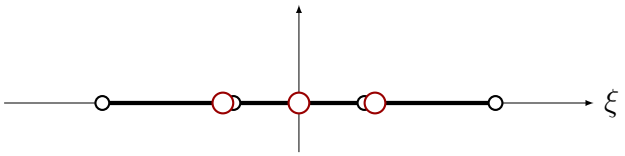
Gauss points	Weights
0	2

Order 3



Gauss points	Weights
$-1/\sqrt{3}$	1
$+1/\sqrt{3}$	1

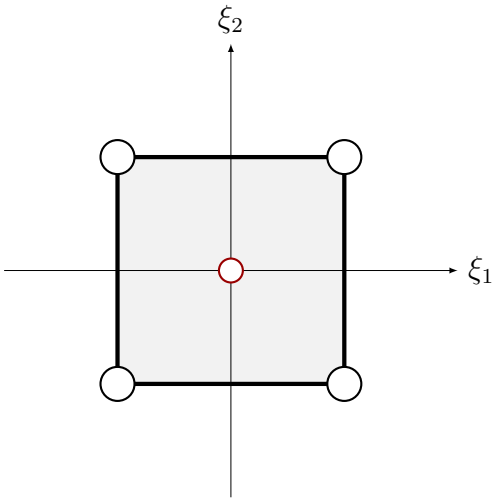
Order 5



Gauss points	Weights
0	$8/9$
$+\sqrt{3/5}$	$5/9$
$-\sqrt{3/5}$	$5/9$

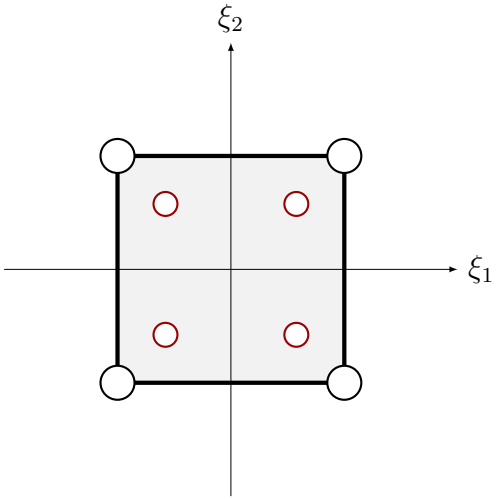
Gauss-Legendre points and weights in 2D

Order 1 - quadrilateral



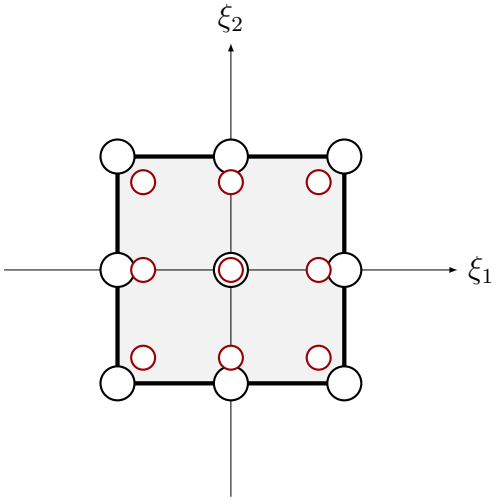
Gauss points	Weights
$(0, 0)$	2

Order 3 - quadrilateral



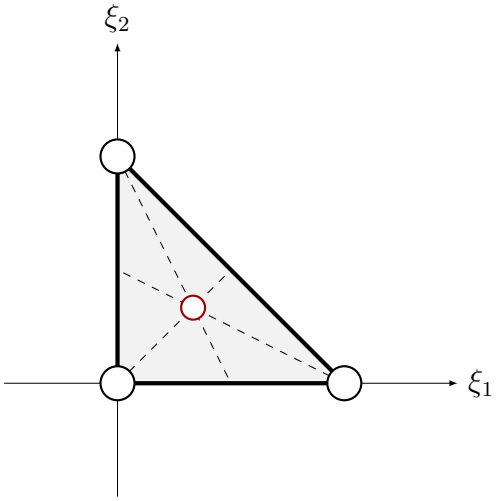
Gauss points	Weights
$(-1/\sqrt{3}, -1/\sqrt{3})$	1
$(-1/\sqrt{3}, +1/\sqrt{3})$	1
$(+1/\sqrt{3}, +1/\sqrt{3})$	1
$(+1/\sqrt{3}, +1/\sqrt{3})$	1

Order 5 - quadrilateral



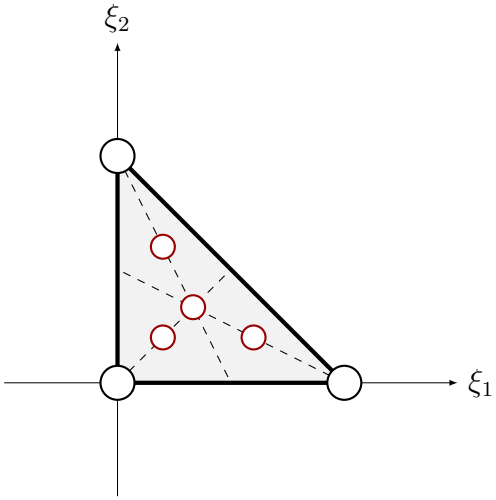
Gauss points	Weights
$(-\sqrt{3/5}, -\sqrt{3/5})$	25/81
$(-\sqrt{3/5}, +\sqrt{3/5})$	25/81
$(+\sqrt{3/5}, -\sqrt{3/5})$	25/81
$(+\sqrt{3/5}, +\sqrt{3/5})$	25/81
$(-\sqrt{3/5}, 0)$	40/81
$(+\sqrt{3/5}, 0)$	40/81
$(0, -\sqrt{3/5})$	40/81
$(0, +\sqrt{3/5})$	40/81
$(0, 0)$	64/81

Order 1 - triangular



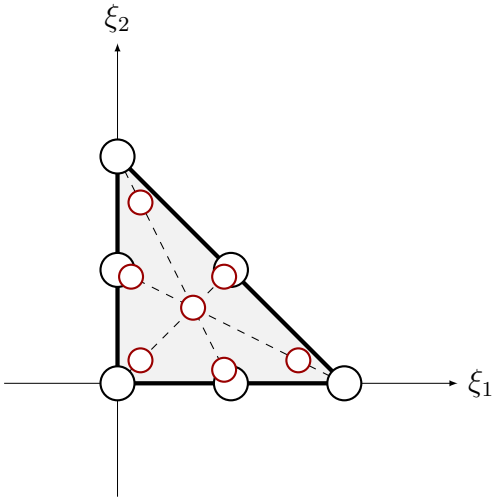
Gauss points	Weights
$(1/3, 1/3)$	$1/2$

Order 3 - triangular



Gauss points	Weights
$(1/5, 1/5)$	$25/96$
$(3/5, 1/5)$	$25/96$
$(1/5, 3/5)$	$25/96$
$(1/3, 1/3)$	$-9/32$

Order 5 - triangular



Gauss points	Weights
$\left(\frac{1905632}{18814273}, \frac{1905632}{18814273}\right)$	$\frac{55504499}{881449264}$
$\left(\frac{395553083}{496036741}, \frac{1905632}{18814273}\right)$	$\frac{55504499}{881449264}$
$\left(\frac{1905632}{18814273}, \frac{395553083}{496036741}\right)$	$\frac{55504499}{881449264}$
$\left(\frac{335541047}{713701395}, \frac{34286004}{574152281}\right)$	$\frac{65172071}{984515851}$
$\left(\frac{335541047}{713701395}, \frac{335541047}{713701395}\right)$	$\frac{65172071}{984515851}$
$\left(\frac{34286004}{574152281}, \frac{335541047}{713701395}\right)$	$\frac{65172071}{984515851}$